Huiseok Moon

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RESEARCH INTERESTS

Assistive robotics and physical human–robot interaction, grounded in clinically validated real-time intent detection and sensor-based assistive control (EMG, IMU). Current focus: physics-based modeling and simulation, model-based control, and multimodal AI for understanding and predicting human motion in wearable robotics.

Keywords: Wearable Robotics • Assistive Robotics • Multimodal Learning • Nonlinear Control

EDUCATION

Ph.D. in Signal, Image and Control — **Nov 2019 – Jan 2024**
University Paris-Est Créteil — Advisor: Dr. Samer Mohammed — *Créteil, France*
Thesis: *Human-in-the-loop Paradigms for Mobility Assistance Using Powered Exoskeleton*

M.S. in Mechanical Engineering — **Mar 2017 – Feb 2019**
Sogang University — Advisor: Dr. Kyoungchul Kong — *Seoul, South Korea*
Thesis: *Control Algorithm of a Robotic Leg for Reducing Interaction Force Between a Load and the Human Body*
Graduate Research Fellowship, Ilwun Science and Technology Foundation (2017–2019)

B.S. in Robotics (Intelligence System) — **Mar 2010 – Feb 2017**
Kwangwoon University — *Seoul, South Korea*
Thesis: *Research on Personal Mobility Using Ball Balancing Robot*

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher — **Jul 2026 – Sep 2026**
Université Paris-Est Créteil — *Vitry-sur-Seine, France*

- Designed a intention detection algorithm within dynamical flow of the latent dynamics.
- Developed an energy-efficient adaptive FES control strategy based on surface EMG signals, reducing stimulation energy consumption by 30% while mitigating muscle fatigue in clinical trials.
- Implemented a 7-IMU multi-sensor fusion framework extracting kinematic metrics for lower-limb joint analysis and Parkinson's disease diagnosis in a portable clinical device.

Postdoctoral Researcher — **Sep 2025 – Mar 2026**
Korea Advanced Institute of Science & Technology (KAIST) — *Daejeon, South Korea*

- Developed physics-informed AI models for human motion intention inference, leveraging motion-primitive representations of wearable IMU signals.

Research Engineer — **Sep 2023 – Aug 2025**
Université Paris-Est Créteil — *Vitry-sur-Seine, France*

- Built the LSTM-based gait mode recognition pipeline (98% accuracy) underpinning real-time control of actuated ankle-foot orthoses.
- Devised an adaptive EMG-driven FES control strategy, cutting stimulation energy by 30% and reducing muscle fatigue in clinical validation.

ATER (Teaching & Research Assistant) — **Nov 2022 – Aug 2023**
Université Paris-Est Créteil — *Vitry-sur-Seine, France*

- Delivered undergraduate lectures and labs on wearable sensor integration (EMG, IMU, FSR) and biomechanical signal processing in Python and MATLAB.

Research Assistant — **Nov 2019 – Oct 2022**
Université Paris-Est Créteil — *Vitry-sur-Seine, France*

- Developed nonlinear control architectures (impedance control, reference tracking) for human–exoskeleton interaction, applied to lower-limb exoskeletons and sit-to-stand assistance robots.
- Constructed a real-time activity recognition pipeline combining Random Forests and deep learning with Kalman-filtered IMU features for intent-based mobility assistance control.

Research Assistant — **Mar 2017 – Feb 2019**
Sogang University — *Seoul, South Korea*

- Designed interaction force control strategies for lower-limb exoskeletons to augment strength and endurance in load-carrying tasks.

PUBLICATIONS

Peer-Reviewed Journal Articles

[J1] O. Bey, M. Chemachema, R. Jradi, **H. Moon**, H. Rifai, Y. Amirat, and S. Mohammed, "Non-model-based finite-time adaptive neural output feedback control for an active ankle–foot orthosis," *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 14103–14119, 2026. [DOI]

[J2] **H. Moon**, O. Bey, A. Boubezoul, L. Oukhellou, and S. Mohammed, "Real-time LSTM-driven dynamic gait mode detection for enhanced control of actuated ankle–foot orthosis," *IEEE Transactions on Robotics*, vol. 41, pp. 4794–4809, 2025. [DOI]

[J3] G. Khodabandelou, **H. Moon**, Y. Amirat, and S. Mohammed, "A fuzzy convolutional attention-based GRU network for human activity recognition," *Engineering Applications of Artificial Intelligence*, vol. 118, Art. no. 105702, 2023. [DOI]

[J4] **H. Moon**, R. Maiti, K. Das Sharma, Y. Amirat, P. Siarry, and S. Mohammed, "Hybrid half-Gaussian selectively adaptive fuzzy control of an actuated ankle–foot orthosis," *IEEE Robotics and Automation Letters*, vol. 7, pp. 9635–9642, 2022. Presented at ICRA 2023 (oral). [DOI]

[J5] I. Jammeli, A. Chemori, **H. Moon**, S. Elloumi, and S. Mohammed, "An assistive explicit model predictive control framework for a knee rehabilitation exoskeleton," *IEEE/ASME Transactions on Mechatronics*, vol. 27, pp. 3636–3647, 2022. [DOI]

[J6] W. Huo, **H. Moon**, M. A. Alouane, V. Bonnet, J. Huang, Y. Amirat, R. Vaidyanathan, and S. Mohammed, "Impedance modulation control of a lower-limb exoskeleton to assist sit-to-stand movements," *IEEE Transactions on Robotics*, vol. 38, pp. 1230–1249, 2022. [DOI]

Submitted Manuscripts

[U1] X. Luo*, **H. Moon***†, J.-M. Gracies, W. Huo†, and S. Mohammed†, "Knee-guided phase–magnitude decoupled adaptive functional electrical stimulation for drop-foot correction," submitted to *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2026.

[U2] L. Marusu, C. Gault-Colas, **H. Moon**, S. Mohammed, and J.-M. Gracies, "Changes in walking speed, step length, and lower-limb kinematics over eight weeks of guided self-rehabilitation contracts in hemiparetic patients: A prospective cohort study," submitted to *Journal of Neural Engineering*, 2026.

* Equal contribution; † Corresponding author.

Peer-Reviewed Conference Proceedings

[C1] O. Bey, R. Jradi, **H. Moon**, H. Rifai, K. Das Sharma, Y. Amirat, and S. Mohammed, "A novel funnel-based L_1 adaptive fuzzy approach for the control of an actuated ankle–foot orthosis," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2024. [DOI]

[C2] **H. Moon**, A. Boubezoul, L. Oukhellou, Y. Amirat, and S. Mohammed, "Online human intention detection through machine-learning based algorithm for the control of lower-limbs wearable robot," in *Proceedings of the IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2022. [DOI]

[C3] P.-G. Jung, W. Huo, **H. Moon**, Y. Amirat, and S. Mohammed, "A novel gait phase detection algorithm for foot drop correction through optimal hybrid FES–orthosis assistance," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2021. [DOI]

ACADEMIC SERVICE AND TEACHING

Peer Review IEEE Transactions on Robotics, IROS, ICRA, Engineering Applications of Artificial Intelligence, Mechatronics, and others

Invited Talks REHAssist Visiting Seminar, EPFL, Switzerland (2025); LISSI–BIOTN Joint Seminar, UPEC, France (2024); LISSI Research Seminars, UPEC, France, 4 sessions (2020–2023)

Volunteering IFAC World Congress, volunteer (2025); IUT Research Day, poster exhibition (2024); Fête de la Science, public outreach exhibitor (2023); Korea Defense Science & Tech Fair, exhibitor (2017)

Mentoring M.S. students: L. Marusu (2025), A. A. Benhamlaoui (2024), D. Plet (2024), W. Legendre (2023), J. M. Fadous (2022), R. Ravaka & M. Diaby (2021) — topics spanning multi-sensor gait analysis, AI-based gait event detection, Parkinson's diagnostics, and assistive robot control

SKILLS

Programming Python, C, C++, MATLAB, LabVIEW, SolidWorks

Frameworks PyTorch, TensorFlow, Simulink

Expertise Real-time ML deployment, multimodal sensor fusion, embedded systems, adaptive & impedance control, human–robot interaction, wearable robotics, signal processing

Languages Korean (native), English (fluent), French (basic)

REFERENCES

Dr. Samer Mohammed
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